

Module 1 — The Regulatory Change Awareness Gap

A Multi-Stage, Multilingual Investigation of Information Lag Between Regulatory Publication and SME Awareness in Sri Lanka

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1. Module Overview

This module is one of four investigations into the underlying information barriers driving regulatory non-compliance among Sri Lankan Small and Medium Enterprises (SMEs). While the other three modules study what SMEs *understand* (Module 2), what predicts their *risk of failure* (Module 3), and how *misinformation circulates* among them (Module 4), this module addresses a more upstream question:

By the time a Sri Lankan SME owner is aware that a regulation has changed, how late are they?

The question sounds simple. It has never been answered for Sri Lanka. There is no published measurement of the time elapsed between a regulatory change being gazetted by the Government of Sri Lanka and the moment a typical SME owner becomes aware of that change. There is no published ranking of which communication channel — official portal, news media, accountant networks, social media, peer-to-peer messaging — delivers regulatory information fastest to SMEs. There is no classifier capable of automatically categorising Sri Lankan regulatory text into actionable SME-relevant change types in Sinhala, Tamil, and English.

This chapter establishes why those gaps matter, surveys the literature that surrounds (but does not fill) them, articulates the novel contributions this module proposes, and details a methodologically defensible plan for empirical investigation.

2. Problem Statement

When the Government of Sri Lanka changes a regulation that affects SMEs — a VAT rate adjustment, a new invoice format, a revised APIT withholding obligation, an updated EPF/ETF contribution rule — that change passes through a long, fragmented, and largely uninstrumented information chain before it reaches the SME owner who must act on it.

The chain typically looks like this:

Stage	What happens	Who is responsible
T ₀	Regulation is drafted and published in the <i>Government Gazette</i>	Government Press / line ministry
T ₁	Change is reflected on the responsible authority's portal (IRD, Department of Labour, eROC, EPF/ETF)	The respective regulator
T ₂	Change is covered by news media (Daily FT, LBO, Daily Mirror, Sinhala/Tamil dailies)	Independent media
T ₃	Change is interpreted and circulated by intermediaries (accountants, chambers, NEDA, professional bodies)	Private intermediaries
T ₄	Change is propagated through peer networks (WhatsApp groups, Facebook business groups, sector forums)	Informal networks
T ₅	The individual SME owner becomes aware of the change	The SME
T ₆	The SME understands what action to take	The SME (often with paid help)

Each transition introduces lag. Each transition introduces fidelity loss. Each transition is uninstrumented. The cumulative effect is that SMEs frequently learn of compliance obligations *after* the deadline to comply has elapsed — at which point penalties, interest, and reputational damage have already accrued.

The fact that this lag exists is not in dispute; the Central Bank of Sri Lanka's 2021 Annual Report (Box 4) and the Ministry of Industry and Commerce's National SME Policy Framework both acknowledge that information access is a structural constraint on Sri Lankan SMEs. What has *not* been done is the empirical work: measuring the lag, decomposing it across the chain, identifying which channels are bottlenecks, and quantifying which SME segments are most affected. **Without measurement there can be no targeted intervention.** This module supplies the measurement.

3. Literature Review

3.1 Regulatory Information Diffusion in Developed Economies

The body of work on how regulatory information reaches firms is concentrated in OECD economies. Studies by the OECD's Regulatory Policy Outlook series document how member states use centralised "one-stop" portals, structured stakeholder consultation, and multi-channel

dissemination to shorten the time between regulatory enactment and firm awareness. Slemrod (2019), in his comprehensive review of tax compliance and enforcement in the *Journal of Economic Literature*, treats taxpayer awareness as an assumed input to compliance models — a reasonable assumption in well-instrumented economies but a significant blind spot when transposed to settings where awareness itself is the bottleneck.

The implicit assumption across this literature is that the regulator-to-firm information channel is short, formal, and high-fidelity. None of the cited works measure end-to-end diffusion lag for a developing economy, and none disaggregate the diffusion path into the multi-stage cascade described in §2 above.

3.2 SME Compliance Burden Research

A separate strand of literature examines the disproportionate compliance burden borne by SMEs (e.g., OECD, 2018; World Bank *Doing Business* reports). This work documents *that* SMEs struggle with compliance — quantifying the time and cost in person-hours and currency — but it largely treats the cause as procedural complexity (too many forms, too many portals) rather than informational latency (the firm did not know in time). The distinction matters: a procedural fix (simpler forms) does not solve an informational problem (the firm learned about the form too late). This module argues that informational latency is a distinct, measurable, and addressable category of compliance burden that has been under-specified in the existing burden literature.

3.3 Information Diffusion Theory

Three theoretical traditions are directly relevant:

Rogers' Diffusion of Innovations theory (Rogers, 2003) describes how new ideas, practices, and information spread through a social system over time. Rogers identifies five stages — *knowledge*, *persuasion*, *decision*, *implementation*, and *confirmation* — and emphasises that the *knowledge* stage is gated by communication channels, social system characteristics, and the perceived attributes of the innovation. While Rogers' framework has been applied extensively to technology adoption, agricultural extension, and public health interventions, its application to *regulatory* information diffusion is rare, and its application to regulatory diffusion in a multilingual developing economy is, to the best of our knowledge, absent from the published literature.

Agenda-Setting Theory (McCombs & Shaw, 1972) provides the framework for understanding the role of mass media in transferring salience from the institutional agenda to the public agenda. Applied to this module, it predicts that the news media (T_2) is a critical bottleneck: regulatory changes that are not picked up by media will diffuse far more slowly to SMEs than those that are.

The Two-Step Flow Model (Lazarsfeld, Berelson & Gaudet, 1944; Katz, 1957) holds that information flows from media to opinion leaders, and from opinion leaders to the wider population. In the SME context, the "opinion leaders" are accountants, chamber officials, NEDA

representatives, and prominent professional voices on social media. This model predicts that the lag between T_2 (media coverage) and T_5 (SME awareness) is substantially mediated by these intermediaries, and that the choice of which intermediaries an SME is connected to is a strong predictor of awareness lag.

The Knowledge Gap Hypothesis (Tichenor, Donohue & Olien, 1970) further predicts that as the information flow into a social system increases, segments with higher socioeconomic status acquire information at a faster rate, *widening* rather than narrowing the gap between segments. Applied here, it predicts that larger and more urban SMEs will have systematically shorter awareness lags than smaller, rural, and minority-language SMEs — a hypothesis this module is positioned to test directly.

3.4 Multilingual Public Communication

Sri Lanka is constitutionally trilingual (English, Sinhala, Tamil), but the regulatory communication infrastructure is overwhelmingly anglophone. Herath (2024), cited in the group's parent proposal, documents that local-language access is a recognised barrier to SME empowerment in Sri Lanka. However, Herath's work is qualitative and policy-oriented; it does not measure the *temporal* effect of language asymmetry on regulatory awareness. There is, more broadly, no published study that quantifies how the language in which a regulation is first published affects how quickly it reaches SMEs operating primarily in another language. This module treats language as a measurable variable in the diffusion model.

3.5 NLP for Regulatory and Legal Text

The natural-language processing community has produced substantial work on legal and regulatory text classification — primarily for English-language jurisdictions (US federal regulations, EU directives, UK statutes). Multilingual transformer models such as mBERT (Devlin et al., 2019) and XLM-RoBERTa (Conneau et al., 2020) have demonstrated strong cross-lingual transfer for South Asian languages on general-domain tasks, but their performance on Sinhala and Tamil regulatory text is undocumented. There is no published Sri Lankan regulatory text corpus, no benchmark for Sri Lankan regulatory classification, and no fine-tuned multilingual model for Sri Lankan regulatory change categorisation. This module produces all three.

3.6 The Sri Lankan Regulatory Communication Landscape

Sri Lanka's regulatory communication infrastructure is fragmented across multiple authorities — the Inland Revenue Department (IRD), the Department of the Registrar of Companies (eROC), the Department of Labour (for EPF), the Employees' Trust Fund (ETF) Board, and the Customs Department, among others. Each authority maintains its own portal, its own gazette tracking practice, and its own communication cadence. There is no unified regulatory feed. The Central Bank of Sri Lanka (2022) and the National SME Policy Framework (Ministry of Industry

and Commerce, 2017) both identify this fragmentation as a structural challenge but neither quantifies its temporal impact on SMEs. This module supplies that quantification.

4. The Identified Research Gap

The literature reviewed in §3 surrounds the problem but does not fill it. We articulate six concrete, defensible gaps that this module is designed to close.

4.1 The Geographic Gap

Empirical measurements of regulatory information diffusion lag are concentrated in OECD economies, where centralised portals and structured communication shorten the diffusion path. No published study measures end-to-end regulatory awareness lag for any South Asian economy, and none for Sri Lanka specifically. The diffusion patterns of a fragmented, trilingual, developing-economy regulatory landscape are absent from the empirical record.

4.2 The Multi-Stage Cascade Measurement Gap

Existing diffusion studies typically measure a single transition — usually publication-to-awareness, treated as a black box. This module decomposes diffusion into a six-stage cascade ($T_0 \rightarrow T_6$) and measures lag at each stage independently. The decomposition is itself a methodological contribution: it makes the *bottleneck* visible rather than only the *total delay*, which has direct implications for where intervention will have the greatest marginal impact.

4.3 The Channel Effectiveness Gap

While studies of SME information channels exist, they are predominantly *preference-based* — asking SMEs which channels they prefer, trust, or use. None systematically rank channels by *measured time-to-awareness* for regulatory information specifically. This module produces the first such ranking for Sri Lanka, distinguishing official, media, intermediary, and informal channels by their empirical delivery speed.

4.4 The Multilingual Diffusion Gap

The effect of language on regulatory information diffusion timing is unmeasured. In a country where regulations are gazetted primarily in English and Sinhala but where many SME owners operate in Tamil or in Sinhala-only contexts, the language differential is hypothesised to introduce additional lag — but the magnitude has never been quantified. This module treats language as a measured variable.

4.5 The Regulatory Granularity Gap

Existing diffusion studies treat regulations as a monolithic category. In practice, a change in tax *rate*, a change in invoice *format*, a new *deadline*, and a new *procedural* requirement may diffuse at very different speeds — because they generate different levels of media attention, different

intermediary interest, and different SME priority. No published work disaggregates lag by regulatory change *type*. This module does.

4.6 The Technical / NLP Gap

There is no Sri Lankan regulatory text corpus, no benchmark dataset for Sri Lankan regulatory classification, and no fine-tuned multilingual NLP model for categorising Sri Lankan regulatory changes by SME relevance. Without such tools, scaling the analysis beyond a hand-coded sample is infeasible. This module constructs the corpus, the benchmark, and the classifier as deliverables in their own right.

5. Research Questions

5.1 Primary Research Question

What is the information lag between the publication of a regulatory change in the Government Gazette of Sri Lanka and the awareness of that change by SME owners — and how does this lag vary across regulatory channels, regulatory change types, languages, and SME characteristics?

5.2 Secondary Research Questions

The primary question is decomposed into five secondary questions, each individually answerable and individually contributing to the overall finding:

- **SQ1 (Cascade):** What is the measured lag between each consecutive stage of the regulatory information cascade ($T_0 \rightarrow T_1, T_1 \rightarrow T_2, T_2 \rightarrow T_3, T_3 \rightarrow T_4, T_4 \rightarrow T_5$) for Sri Lankan regulatory changes?
 - **SQ2 (Channel ranking):** Among the channels available to SMEs, which channel delivers regulatory information with the lowest median time-to-awareness?
 - **SQ3 (Type variation):** Do different categories of regulatory change (rate change, format change, deadline change, procedural change, threshold change) exhibit statistically different diffusion timings?
 - **SQ4 (Demographic variation):** Does awareness lag vary systematically with SME characteristics — sector, employee count, annual turnover, geographic region, primary operating language, and owner education level?
 - **SQ5 (Predictability):** Can the awareness lag for a future regulatory change be predicted from features of the change itself (its type, its language of first publication, its issuing authority, its early media coverage)?
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6. Theoretical Framework

This module synthesises the four theoretical traditions surveyed in §3.3 into a single integrated framework.

The **Rogers diffusion model** supplies the temporal stages. The **Agenda-Setting model** supplies the prediction that media attention at T_2 disproportionately shapes downstream awareness. The **Two-Step Flow model** supplies the prediction that intermediary networks (T_3) are the dominant determinant of SME awareness rather than direct media exposure. The **Knowledge Gap hypothesis** supplies the prediction that lag will not be uniformly distributed across SMEs — that larger, urban, English-operating SMEs will have systematically shorter lags, and that the introduction of new regulatory information may *widen* rather than narrow inter-SME information disparities.

These frameworks generate testable hypotheses:

- **H1 (Cascade hypothesis):** Lag is non-uniformly distributed across the six-stage cascade; one or two stages will dominate the total delay.
- **H2 (Intermediary dominance hypothesis):** SMEs whose primary information source is an intermediary (accountant, chamber, professional body) will have lower median awareness lag than SMEs whose primary source is direct media or official portals.
- **H3 (Type heterogeneity hypothesis):** Rate changes will diffuse faster than format and procedural changes, due to higher media salience.
- **H4 (Language asymmetry hypothesis):** SMEs operating primarily in Tamil will exhibit longer awareness lags than Sinhala-operating SMEs, who will in turn exhibit longer lags than English-operating SMEs, for changes first published in English.
- **H5 (Knowledge-gap-widening hypothesis):** The variance of awareness lag across the SME population will be larger for high-volume regulatory periods than for low-volume periods, consistent with the Tichenor et al. prediction.

These hypotheses are pre-registered to discipline the analysis and to ensure the contribution is genuine measurement rather than post-hoc storytelling.

7. Novelty and Contributions

The novelty of this module is articulated across five distinct, defensible axes. Each is presented with a precise statement of what is new and what existing work it exceeds.

7.1 Methodological Novelty — The Six-Stage Cascade Lag Framework

Previous work has measured regulatory diffusion as a single end-to-end interval. This module

introduces a **six-stage cascade decomposition** (Gazette → Portal → Media → Intermediary → Peer Network → SME Awareness → SME Comprehension) with independent measurement at each stage. The framework itself is a contribution: it can be reapplied to any other developing economy with a fragmented regulatory landscape, and it transforms the question from "how long is the lag?" to "*where in the chain* is the lag concentrated?" — which is the question intervention design actually requires.

7.2 Empirical Novelty — First Measured Lag for Sri Lanka

This module produces the first published measurement of regulatory awareness lag for Sri Lanka. The headline numbers — median end-to-end lag, stage-wise lag distribution, channel-ranked lag, type-stratified lag, language-stratified lag — do not currently exist in the public record. They exist after this module.

7.3 Dataset Novelty — The Sri Lankan Regulatory Change Timeline Corpus

The module produces a public, structured, time-stamped corpus of regulatory changes affecting Sri Lankan SMEs from 2018 to 2026, with each change linked to its first portal publication date, its first media coverage date(s), and (for a sampled subset) its measured SME awareness distribution. The corpus is the first of its kind for Sri Lanka and is itself a reusable research artefact for subsequent work in regulatory communication, compliance economics, and computational legal studies.

7.4 Technical Novelty — The Trilingual Sri Lankan Regulatory Classifier

The module produces a fine-tuned multilingual classifier (XLM-R based, with TF-IDF and mBERT baselines for comparison) capable of categorising Sri Lankan regulatory text in Sinhala, Tamil, and English by (a) issuing authority, (b) regulatory change type, and (c) SME relevance level. This is the first such classifier for Sri Lankan regulatory text and contributes a labelled training set, evaluation benchmark, and reproducible model artefact to the under-resourced South Asian NLP ecosystem.

7.5 Contextual Novelty — The Trilingual Diffusion Model

The module produces the first diffusion model that explicitly incorporates language as a measured variable in regulatory information cascades. The empirical results — whether or not Tamil-operating SMEs experience longer lags, by how much, and which stage of the cascade is the bottleneck — directly inform language policy in regulatory communication for Sri Lanka and generalise to other multilingual developing economies (India, Pakistan, Bangladesh, Nepal, Indonesia, the Philippines).

7.6 Summary Novelty Table

Contribution	Type	Why it is new
Six-stage cascade lag framework	Methodology	No multi-stage cascade decomposition exists for regulatory diffusion
Measured awareness lag for Sri Lanka	Empirical finding	Has never been measured
Sri Lankan regulatory change timeline corpus	Dataset	Does not exist in any public repository
Trilingual Sri Lankan regulatory classifier	Technical artefact	No such model exists
Channel effectiveness ranking	Empirical finding	No measured ranking for SME regulatory information
Type-stratified diffusion analysis	Empirical finding	Existing work treats regulation as monolithic
Language-stratified diffusion analysis	Empirical finding	Language has never been treated as a measured variable in this context

8. Methodology

8.1 Research Design

A **convergent mixed-methods design** is employed. Quantitative components (corpus construction, lag measurement, NLP classification, statistical analysis) and qualitative components (open-text survey responses, intermediary interviews) are collected in parallel and integrated at the analysis stage. The design is pre-registered: the hypotheses listed in §6 are committed in advance of analysis and the analysis plan is fixed before the data is examined, to prevent post-hoc selection of favourable results.

8.2 Data Sources and Collection Protocols

#	Source	Data extracted	Method	Time horizon
1	<i>Government Gazette of Sri</i>	Publication date,	Automated PDF retrieval;	2018-01-

#	Source	Data extracted	Method	Time horizon
	<i>Lanka</i> (documents.gov.lk)	issuing authority, full text of regulatory notices affecting SMEs	PyMuPDF / pdfplumber for text extraction; manual verification of date stamps	01 to 2026-06-30
2	IRD, eROC, EPF/ETF, Department of Labour portals	First-appearance date of each gazetted change on the responsible authority's portal	Custom Scrapy crawlers with versioned snapshots; cross-validated against web.archive.org snapshots	2018-01-01 to 2026-06-30
3	News archives — <i>Daily FT, LBO, Daily Mirror, Sunday Times, Lankadeepa, Dinamina, Veerakesari, Thinakkural</i>	First news article date covering each gazetted change; subsequent coverage volume	Search-based archive retrieval with regulatory change keywords; manual verification of first-coverage attribution	2018-01-01 to 2026-06-30
4	SME Awareness Survey	Self-reported date of first awareness of each surveyed regulatory change; channel through which awareness occurred; SME demographic data	Structured online survey (Google Forms) distributed via the Ceylon Chamber of Commerce, the National Chamber of Commerce of Sri Lanka, NEDA networks, and LinkedIn outreach; trilingual instrument (English, Sinhala, Tamil)	Cross-sectional, 2026
5	Intermediary interviews	Qualitative narrative of how regulatory information reaches and is interpreted by intermediaries	Semi-structured interviews with 10–12 chartered accountants, chamber officials, and tax practitioners	2026 Q2

8.3 Sampling Strategy and Sample Size

The SME survey targets a stratified random sample drawn from the publicly listed Sri Lankan SME population. Strata are defined along three dimensions:

- **Sector** (manufacturing, services, trade, agriculture, construction)
- **Size band** (micro: 1–10 employees; small: 11–50; medium: 51–300)

- **Geographic region** (Western Province; other provinces aggregated)

Sample size is calculated using Cochran's formula for proportions, assuming a 95 % confidence level, a 5 % margin of error, and a conservative response variance of 0.5. The unadjusted target is **n = 384**; allowing for an estimated 30 % completion rate, the gross outreach target is **≈ 1 280 SMEs**. To ensure adequate cell sizes for stratified analysis (H4, H5), the minimum target per stratum is 25, requiring a **total realised sample of n ≥ 400 with stratum oversampling for under-represented cells** (Tamil-operating, rural, agriculture-sector). The realistic working target for this module's primary analysis is **n = 250–400** completed responses; secondary analyses contingent on sufficient cell sizes are clearly flagged in advance.

8.4 Survey Instrument Design

The instrument has five sections:

1. **SME profile** — sector, size, location, age, primary operating language, owner education
2. **Awareness probes** — for each of 6–8 specific recent regulatory changes (e.g., 2024 VAT rate change to 18 %, 2026 VAT invoice format change, APIT withholding obligation), the respondent indicates: (a) whether they were aware before the effective date, (b) approximate date of first awareness, (c) channel of first awareness, (d) whether they took action on time
3. **Channel inventory** — for each of ~10 channels, frequency of use and trust rating
4. **Open-text** — qualitative narrative of the respondent's most recent compliance information experience
5. **Consent and ethics** — informed consent, anonymity affirmation, contact for follow-up (optional)

The instrument is professionally translated into Sinhala and Tamil and back-translated for fidelity verification. It is piloted with 15–20 respondents prior to full deployment. Response burden is targeted at 12–15 minutes.

8.5 Gazette and News Corpus Construction

The gazette corpus is constructed by automated retrieval of all gazette PDFs from `documents.gov.lk` for the period 2018-01-01 to 2026-06-30, followed by:

- Text extraction (PyMuPDF primary, pdfplumber fallback for malformed PDFs)
- Section segmentation into individual regulatory notices
- Filtering to SME-relevant notices (issued by IRD, eROC, EPF, ETF, Labour, Customs)
- Manual sampling and verification of extraction fidelity (target: ≥ 95 % accuracy on a 200-document sample)

8.6 NLP Classifier Development

A multilingual classifier is developed to categorise gazette notices by:

- **Issuing authority** (IRD / eROC / EPF / ETF / Labour / other)
- **Change type** (rate change / threshold change / format change / deadline change / procedural change / new obligation / repeal)
- **SME relevance** (high / medium / low / not applicable)

Models compared:

1. **TF-IDF + Logistic Regression** (baseline, monolingual per language)
2. **mBERT fine-tuned** on the labelled corpus
3. **XLM-RoBERTa fine-tuned** on the labelled corpus

Training data: A labelled subset of 1 500–2 000 gazette notices, hand-coded by the researcher and a second annotator, with inter-annotator agreement measured via **Cohen's κ** (target ≥ 0.75 for substantial agreement).

Evaluation: Stratified 5-fold cross-validation on the labelled set; held-out test set of 300 notices; metrics reported as macro F1, per-class precision/recall, and per-language F1 to detect cross-lingual performance asymmetry.

8.7 Lag Measurement Protocol

For each regulatory change in the corpus, the following timestamps are constructed:

- T_0 = gazette publication date (from PDF metadata, verified against gazette index)
- T_1 = first portal-appearance date (from snapshot crawl; verified against web.archive.org)
- T_2 = first news-coverage date (from archive search; verified)
- T_3 = first intermediary acknowledgment date (from interview-corroborated data; sampled)
- T_4 = first peer-network appearance (from Module 4 social-media corpus collaboration)
- T_5 = SME awareness date (from survey, distribution rather than point estimate)

Lag at each stage is computed as the difference between consecutive timestamps. End-to-end lag is $T_5 - T_0$.

8.8 Statistical Analysis Plan

The analysis plan is fixed in advance:

- **Descriptive:** median, IQR, and full distribution of lag at each stage; box-plot decomposition of cascade
- **Survival analysis:** Kaplan-Meier curves for time-to-awareness, stratified by channel, change type, language, and SME segment
- **Cox proportional hazards** regression with predictors {channel, type, language, sector, size, region, owner education}; hazard ratios reported with 95 % CIs
- **ANOVA / Kruskal-Wallis** for type-stratified and language-stratified comparisons (H3, H4)
- **Levene's test** and variance-ratio analysis for the knowledge-gap-widening hypothesis (H5)
- **Multiple-comparison correction** via Benjamini-Hochberg FDR at $\alpha = 0.05$

All code, anonymised data, and analysis notebooks are committed to a public GitHub repository upon thesis submission.

9. Validation Strategy

The findings are validated along four axes:

1. **Source triangulation.** Each timestamp T_0 - T_4 is constructed from independent sources; discrepancies are reconciled via documented protocol rather than discretion.
 2. **Inter-rater reliability.** All hand-coded data (regulatory categorisation, awareness-channel attribution) is double-coded for a sampled subset, with $\kappa \geq 0.75$ required for the coding scheme to be retained.
 3. **Member checking.** Aggregate findings are presented to a small panel of intermediary stakeholders (chartered accountants, chamber officials) for face-validity assessment before publication.
 4. **Solution validation.** The module's downstream artefact — an automated gazette monitoring and SME alert system — is validated by demonstrating that it delivers regulatory information at a measured time-to-awareness *significantly shorter* than the survey-measured baseline lag. This closes the research loop: the *measured* problem is addressed by an *intervention measured against the same metric*.
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10. Expected Outcomes and Deliverables

#	Deliverable	Form
1	Sri Lankan Regulatory Change Timeline Corpus (2018-2026)	Public dataset (CSV + JSON, with PDF source references)
2	Trilingual regulatory text classifier	HuggingFace-hosted model + reproducible training notebook
3	Cascade lag measurements for SME-relevant regulatory changes	Dataset + statistical results
4	Channel effectiveness ranking	Table + visualisation
5	Type-stratified diffusion analysis	Tables + survival curves
6	Language-stratified diffusion analysis	Tables + survival curves
7	Automated gazette monitor and SME alert system	Working prototype, integrated with the unified Enigmatrix platform
8	Module 1 thesis chapter	Written report (~25-30 pages)
9	Conference / journal paper draft	One paper submission target: a regional conference (e.g., ICTer, NCTM) or a Q3+ journal

11. Risks, Limitations, and Mitigation

Risk	Likelihood	Impact	Mitigation
Survey response rate below target	Medium	High	Multi-channel outreach (Chambers, NEDA, LinkedIn, direct sector outreach); incentive (compliance-summary report for respondents); extended fielding window
Gazette PDFs malformed or unmachine-readable	Medium	Medium	Multiple parser fallback; manual extraction for the SME-critical subset
News archives behind paywall or unsearchable	Medium	Medium	Use of archive.org snapshots; partnership outreach to journalism schools; manual sampling

Risk	Likelihood	Impact	Mitigation
			where automated retrieval fails
Tamil-language sample under-recruited	High	Medium	Targeted outreach to Tamil chambers and trade associations; oversampling and post-stratification weighting
Recall bias in self-reported awareness dates	High	Medium	Anchoring questions to specific verifiable changes; soliciting "month and year" rather than precise dates; treating the data as distributional rather than point-estimate
Classifier under-performance on Tamil text	Medium	Medium	Baseline + mBERT + XLM-R comparison surfaces this; cross-lingual transfer learning; explicit data augmentation for Tamil
Generalisability beyond Sri Lanka	N/A	Low	The framework is positioned as transferable; generalisability is a question for future work, not a limitation of this work

12. Alignment with the Other Three Modules

This module does not stand alone. It feeds the unified Enigmatrix platform along three interfaces:

- → **Module 2 (Knowledge Gap):** The regulatory change timeline corpus and the classifier output feed directly into Module 2's verified knowledge base. When a new regulation is detected by Module 1's monitor, Module 2's RAG knowledge base is updated automatically.
- → **Module 3 (Risk Gap):** When a new regulation is detected, Module 3's risk model re-scores affected SMEs to flag those whose profile makes them most likely to miss the new obligation.
- → **Module 4 (Misinformation Gap):** Module 1's awareness-lag findings provide the temporal hypothesis Module 4 tests directly — that misinformation spikes during the lag window, when accurate information is scarce and informal sources fill the vacuum.

In aggregate, the four modules answer four facets of one research question: **why do Sri Lankan SMEs fail at compliance despite intending to comply?** Module 1's specific contribution to that answer is: *because the information arrives too late, through the wrong channels, in the wrong language, for the wrong segment of the SME population — and we now have measured numbers for each of those words.*

13. Timeline (Module-Specific)

Phase	Period	Module 1 activities
Phase 1 — Problem & Literature	Feb 2026	Literature review consolidation; ethics committee submission for survey; pre-registration of hypotheses
Phase 2 — Data Collection	Mar 2026	Gazette + portal + news scraping; survey instrument finalisation, translation, pilot, and full deployment; intermediary interview recruitment
Phase 3 — Analysis & Modelling	Apr 2026	Corpus annotation; classifier training and evaluation; lag measurement; statistical analysis per pre-registered plan
Phase 4 — Platform Integration	May 2026	Automated gazette monitor integrated with unified platform; alert system deployed for closed-cohort testing
Phase 5 — Validation	Jun 2026	Solution-vs-baseline comparison; member checking with intermediary panel; manuscript draft
Phase 6 — Write-up	Jul-Aug 2026	Thesis chapter; paper submission; viva preparation

14. Summary Statement for the Supervisor

This module measures something that has never been measured in Sri Lanka: how late SMEs are when they finally hear about the regulatory changes that govern their own businesses. It does so with a methodologically defensible six-stage cascade framework, a stratified survey instrument, a triangulated public-data corpus, and a trilingual NLP classifier that does not currently exist in the literature. It produces empirical findings, a reusable dataset, a deployable model, and a validated intervention. The contribution is novel along five axes — methodological, empirical, dataset, technical, and contextual — each individually defensible and collectively constituting a coherent research output suitable for both the final-year thesis and a peer-reviewed publication.

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End of Module 1 Chapter.